Bench exercise: two timbers that lock

Japanese Joinery by Hand · Episode 8. Build the watari-ago 渡り腮 and a half-lap 相欠き from stiff card or foam board. Compare how each sits and what stops each from moving. You only need a craft knife, a steel rule and glue. About 45 minutes.

1 · The lower timber

1. Make a long box from card: 35 mm wide, 40 mm high, about 250 mm long. That's about a third of the episode's 105 × 120 mm timber.
2. Mark a crossing zone in the middle, as long as the upper timber is wide (35 mm).
3. Inside that zone, lower the top by 5 mm along both edges. Leave a strip about 12 mm wide standing in the middle, running along the timber.

2 · The upper timber

1. Make a second box, 35 mm wide, 60 mm high, about 250 mm long.
2. Across its underside, in the middle, cut a groove 12 mm wide and 5 mm deep, running across the timber from side to side.

3 · Put it together

1. Lay the upper timber across the lower one at right angles and let it drop in. The raised middle goes into the groove.
2. Try to slide it along each timber, then try to lift it.

4 · The half-lap, for comparison

1. Make two boxes 35 × 35 mm. Cut each to half its depth where they cross: one from the top, one from below.
2. Assemble them. Are the tops flush? What stops each piece sliding?

What to notice

- In the watari-ago, both timbers are cut, but shallowly, and they end up at different heights.
- In your model, sliding is blocked both ways, but nothing holds it down. That's why the specification pairs the joint with hardware: a plate bolt, staples, nails or bolts, depending on the use.
- Look at where the cuts are: where one timber rests on the other, not out in the open span.

These proportions are illustrative. No modern specification gives a depth for these cuts. The model follows a dictionary's description (精選版 日本国語大辞典 and the JAANUS glossary); detail drawings for particular uses may differ.